A Nonlinear Granger Causality Learning Approach via Sparse Component-Wise MLP for iEEG-based Epileptogenic Zone Localization

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***Abstract*—Preoperative localization of the epileptogenic zone (EZ) is crucial for patients with drug-resistant epilepsy who require surgical removal of the EZ. In this paper, we propose a nonlinear Granger causality learning approach via sparse component-wise multilayer perceptron (cMLP) for EZ localization. The model takes the historical signals of each intracranial electroencephalography (iEEG) channel as input to learn the nonlinear causal interactions among channels. Component-wise structure separates the influence of each input channel on every output channel, and applying sparsity regularization to the weights of each sub-model allows the identification of significant effective connectivity. By computing the local features of the effective connectivity, we can analyze the possibility of epilepsy in different brain regions and determine the EZ. The proposed approach is evaluated on the iEEG data collected at the Hospital of the University of Pennsylvania (HUP). The results indicate that this approach can accurately identify EZ which is consistent with the clinical annotations. The precision, sensitivity, and F1 Score are 88.9%, 100%, and 0.941, respectively.**

Keywords—Intracranial EEG, formatting, sparsity regularization, effective connectivity, epileptogenic zone localization, component-wise MLP

I. INTRODUCTION

Epilepsy is a common and serious chronic neuropathic condition, affecting approximately 60 million individuals globally [1]. Its onset is caused by abnormal discharges in specific brain regions, which then spread to other normal brain areas and impair their function [2]. Medication is the primary treatment for epilepsy, but about 30–40% of patients do not respond to drugs. Such patients need surgical resection of the EZ, but the efficacy depends on the accurate localization of the EZ before surgery [3]. At present, localization of epileptogenic zone mainly relies on clinicians' subjective analysis of iEEG, which is very time-consuming and prone to human mistake. The non-linear and non-stationary features of EEG signals and the possibility of EEG activity similar to the abnormal discharge form of epileptogenic zone in normal brain areas further increase the difficulty of localization.

iEEG provides EEG signals with high spatial resolution and high signal-to-noise ratio, which can more accurately reflect abnormal discharge activity of the brain, thus becoming an important method for preoperative localization of epileptic foci [4]. The existing methods for epileptogenic zone localization mainly include channel-wise analysis and network-based analysis. Channel-based analysis methods (such as high-frequency oscillation HFOs) can effectively capture abnormal activity in local brain regions and provide reliable information for lesion identification [5]. The method based on brain network analysis reveals the propagation mechanism of epileptogenic activity at the system level by characterizing the connectivity of different brain regions. According to the nature of connection relationships, brain network analysis can be further divided into structural connectivity networks, functional connectivity networks, and effective connectivity networks [6]. Specifically, functional connectivity networks refer to the statistical relationships of specific physiological signals over time, using the synchrony between brain structures to reflect the communication between regions [7]. Effective connectivity network observation and analyzing the working principles and information transmission mechanisms of the brain can reflect the causal effects between brain regions and has greater potential in EZ localization [8]. Previous studies have used linear Granger causality models or frequency-domain methods to construct effective connectivity networks [9]. In the field of deep learning, there has also been corresponding research on model interpretability [10] or on adding neural perturbations to established brain models to construct effective connectivity networks [11]. These methods have made significant progress in revealing information transmission between brain regions and the mechanisms of epileptic seizures.

This paper presents a nonlinear Granger causality learning approach via sparse cMLP for accurate localization of the EZ among patients with drug-resistant epilepsy. The proposed approach constructs effective brain connectivity networks from iEEG signals and subsequently localizes the EZ based on the local features of these networks. Specifically, we utilize the cMLP framework for nonlinear causal modeling to learn nonlinear causal relationships between brain regions from multi-channel iEEG recordings. The component-wise architecture explicitly disentangles the individual contributions of each input channel to each output channel. By imposing sparsity constraints on specific weights within each sub-module of the cMLP, the model automatically selects key causal connections. This process yields a time-varying effective connectivity network that reflects the brain's dynamic causal interactions. The epileptogenicity of each brain region is assessed by evaluating the local features of the connectivity networks, enabling the automated identification of the seizure focus. This approach balances nonlinear modeling capability with interpretability of causal relationships, providing a quantifiable basis for revealing the

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characteristics of abnormal discharges originating from the EZ.

The remainder of the paper is structured as follows. Section 2 presents the methodology used in this approach. In Section 3, we present the application of our approach in localizing EZ in iEEG and discuss the results. Finally, conclusion and future directions are given in Section 4.

II. Method

The overall framework of the proposed approach is shown in Fig. 1, encompassing key steps such as nonlinear Granger causality modeling, sparsity-constrained optimization, dynamic effective connectivity network construction and EZ localization. In the left panel of Fig. 1, a cMLP is employed as a nonlinear Granger causality model, where independent modeling networks capture the dependencies of each output variable separately. The middle panel of Fig. 1 demonstrates that, based on the model prediction error, group loss is applied to specific layer weights to introduce sparsity constraints, and by aggregating the weights, the causal influence of each input channel on the current output channel can be obtained. In the right panel of Fig. 1, the causal relationships from all models are combined to construct an effective connectivity network among brain regions. The sliding-window method is used to capture its dynamic evolution, and the information flow between iEEG channels is computed for each time window to characterize the network dynamics of the channels, enabling channels with sustained abnormal activity to be identified as the epileptogenic zone.

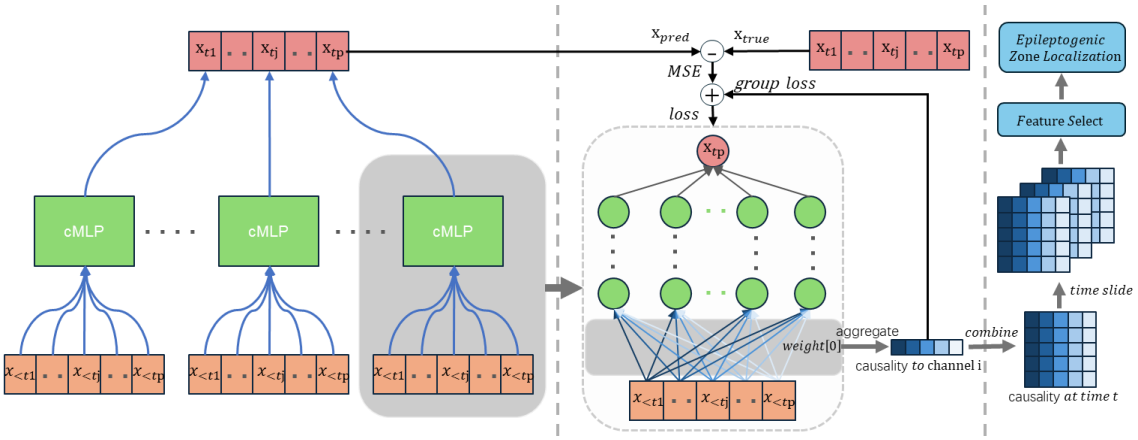


Fig. 1. Overall framework of the proposed method.

A. Modeling Multichannel Nonlinear Effective Connectivity in the Brain

To characterize information transfer among brain regions, this study constructs a multichannel effective connectivity network based on nonlinear Granger causality (GC) analysis using a cMLP model.

1) Effective Connectivity Modeling Based on Nonlinear Granger Causality

The effective connectivity among multichannel brain signals can be characterized using the GC model. Under the linear assumption, GC is typically modeled via a vector autoregressive (VAR) model [12]. Let $x_t \in \mathbb{R}^p$ denote a p-dimensional stationary time series at time t. The VAR model assumes that x_t is a linear combination of its past K lagged vectors:

$$x_t = \sum_{k=1}^{K} \Phi^k x_{t-k} + \varepsilon_t \tag{1}$$

where Φ^k is a $p \times p$ coefficient matrix and ε_t is zero-mean white noise. If $\Phi^k_{i,j} = 0$ for all lags k, channel j is not considered to have a Granger causal effect on channel i.

However, EEG signals exhibit strong nonlinearity and non-stationarity, especially during seizures, making the linear model insufficient to capture complex dynamic interactions between channels. Therefore, a nonlinear Granger causality model is introduced, which assumes that the time series at time t is a nonlinear function of historical channel information:

$$x_t = g(x_{<t1}, \cdots, x_{<tp}) + \varepsilon_t \tag{2}$$

where $x_{<ti} = (\ldots, x_{t-2,i}, x_{t-1,i})$ represents the history of channel i, ε_t is zero-mean white noise, and the function $g(\cdot)$ captures the nonlinear influence of past values on the current value.

To determine whether channel j is a Granger causal source for channel i, we model each output channel i separately:

$$x_{ti} = g_i(x_{<t1}, \cdots, x_{<tp}) + \varepsilon_{ti} \tag{3}$$

If the function $g_i(\cdot)$ is insensitive to the input from channel j, then channel j is not considered a Granger cause of channel i, that is when $x_{<tj} \neq x'_{<tj}$:

$$g_i(x_{<t1}, \cdots, x_{<tj}, \cdots, x_{<tp}) = g_i(x_{<t1}, \cdots, x'_{<tj}, \cdots, x_{<tp}) \tag{4}$$

2) Nonlinear Granger Causality Modeling Using cMLP

To capture the information transfer among brain regions in a nonlinear manner, we adopt a cMLP structure. The cMLP constructs a separate nonlinear prediction model for each output channel i, taking the historical sequences of all channels as input to predict the current value of the channel, thereby learning nonlinear causal relationships between channels.

At time t, the hidden layer vector of the first layer is computed as:

$$h_t^1 = \sigma(\sum_{k=1}^{K} W^1 x_{t-k} + b^1) \tag{5}$$

where W^1 and b^1 are the weight matrix and bias of the first layer, The symbol $\sigma(\cdot)$ represents a nonlinear activation

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function. The hidden layer at layer l is recursively computed as:

$$h_t^l = \sigma(W^l h_t^{l-1} + b^l) \tag{6}$$

The output layer is given by:

$$x_{ti} = g_i\left(x_{<t1}, \cdots, x_{<tp}\right) + \varepsilon_{ti} = W^L h_t^{L-1} + b^L + \varepsilon_{ti} \tag{7}$$

where $W^L$and $b^L$are the parameters of the output layer, and $\varepsilon_{t,i}$is assumed as a white noise with zero mean.

B. Sparsity-Inducing Penalty on Network Weights

The black-box nature of neural networks makes it difficult to directly extract causal relationships. To enhance model interpretability, a sparsity-inducing penalty on weights is introduced to shrink the model coefficients, resulting in a sparse representation where some weights are exactly zero. This mechanism enables the selection of channel-relevant information and facilitates causal interpretation of the learned parameters.

In (6), if all elements in the j-th column of the first-layer weight matrix W^1 for function g_i are zero, denoted as $W^1_{(:,j)} = 0$, then the input channel x_{tj} does not affect either the hidden representation h_t^1 or the output x_{ti}, which corresponds to the situation described in (5), implying that the function $g_i(\cdot)$ is independent of x_{tj}. Consequently, channel j is not a Granger cause of channel i [10].

Therefore, by applying a sparsity-inducing regularization on W^1, the model is encouraged to drive irrelevant column weights toward zero, effectively identifying Granger causal relationships between channels. The overall optimization problem can be formulated as:

$$\min_W \sum_{t=K}^{T} \left(x_{ti} - g_i\left(x_{(t-1):(t-K)}\right)\right)^2 + \lambda \sum_{j=1}^{p} \Omega\left(W_{:j}^1\right) \tag{8}$$

where $\Omega(\cdot)$ is a penalty function that enforces sparsity by shrinking the magnitude of the j-th column of W^1. In this study, a group lasso penalty [13] is used as $\Omega(\cdot)$, which performs shrinkage on the weights of the model's input channels, thereby automatically suppressing irrelevant channels and enhancing the model's causal interpretability:

$$\Omega\left(W_{:j}^1\right) = \left\|W_{:j}^1\right\|_F \tag{9}$$

where $\|\cdot\|_F$ denotes the Frobenius norm. This penalty will uniformly shrink all weights related to input series j. As a result, many columns of W^1 become entirely zero, leaving only those channels that contribute significantly to the model output. Consequently, this regularization mechanism preserves only the most reliable Granger causal relationships among EEG channels.

C. Epileptogenic Zone Localization Using Effective Connectivity Network Features

The causal relationships between EEG channels i and j are characterized through the sparse coefficients obtained from the nonlinear Granger causality model. These coefficients represent the directed information flow between channels and can be used for epileptogenic zone localization. Due to abnormal discharges in epileptic brain regions, the effective connectivity between EEG channels changes continuously over time. To capture these changes, a sliding window approach is applied to segment the iEEG data, enabling time-resolved analysis of effective connectivity patterns. The information flow between channels is obtained within every sliding window. We can use this information to build effective connectivity networks. Graph-theoretical measures are then applied to this network to localize potential epileptogenic zones.

Assuming there are *N* EEG channels, the connectivity between channels are represented by a matrix $C = [c_{ij}] \in \mathbb{R}^{N\times N}$, each c_{ij} denotes the directed causal strength from channel *j* to channel *i*. For each channel *i*, the total information outflow is defined as the total of causal influences from *i* to all other channels, while the total inflow measures the cumulative influence received by channel *i* from all others. Finally, the net information flow is then defined as the difference between the above two:

$$\begin{aligned} outflow_i &= \textstyle\sum_{j=1,\ j\neq i}^{N} c_{ij} \\ inflow_i &= \textstyle\sum_{j=1,\ j\neq i}^{N} c_{ji} \\ netflow_i &= outflow_i - inflow_i \end{aligned} \tag{10}$$

With the above description, we can obtain the local features of each brain effective connectivity network. Outflow reflects a channel's ability to act as a driving source and exert causal influence on other brain regions. If the outflow of a channel is consistently high, it may indicate that this region occupies a central position in the epileptic network, actively initiating the abnormal discharge. Inflow represents the extent to which a channel receives input from other brain regions; The higher inflow may indicate that this region is more susceptible to synchronous drive or passive participation in abnormal activity by the epileptic network. Net flows, calculated as the difference between outflows and inflows, can identify regions that act as "net drivers" or "net recipients" in information exchange. Together, these features characterize the dynamic changes of the brain effective connectivity network during seizures.

To avoid the interference of occasional fluctuations on the localization of EZ, the continuous-time judgment constraint was introduced. First, sort the information flow indicators of the iEEG channels in each time window, and select the channels in the top *K*% as suspected EZ channels. Further calculate the longest consecutive duration and the total number of anomalies in each channel within the continuous time window, and take the channels that exceed the set threshold as the predicted EZ channels.

III. Results and Discussion

A. Dataset

This study utilizes the publicly available OpenNeuro dataset ds004100, originally collected at the Hospital of the University of Pennsylvania (HUP). The dataset comprises de-identified iEEG recordings from 58 individuals diagnosed with drug-resistant epilepsy. These recordings were obtained using a combination of subdural electrodes (ECoG) and stereotactically implanted depth electrodes (sEEG) [14]. All enrolled patients subsequently received the assigned surgical intervention, namely resection or laser ablation. Treatment results were systematically assessed according to the Engel Epilepsy Surgery Outcome Scale. Patients classified as Engel

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Class I are generally free from disabling seizures; consequently, the analysis in this work primarily focuses on data from this specific patient subgroup.

The dataset provides detailed spatial coordinates for each electrode contact, along with clinically annotated epileptogenic channels and records of the specific channels that were either surgically resected or thermally ablated. These comprehensive annotations establish a reliable foundation for the localization and analysis of the epileptogenic zone.

As an example, subject sub-HUP 116 was implanted with sEEG electrodes at a sampling rate of 500 Hz, comprising 52 total channels, among which 50 were usable. The EZ was identified in the RA and RH channels. This patient achieved an Engel Class I postoperative outcome, indicating the absence of disabling seizures. Fig. 2 illustrates a segment of sEEG recordings from the patient's first seizure episode. During data preprocessing, bad channels were first removed. Subsequently, a low-pass filter was applied to suppress high-frequency components above 80 Hz, using a finite impulse response (FIR) filter with a Hann window to reduce ringing artifacts. A notch filter was then applied to suppress power line interference and its second harmonic (60 Hz and 120 Hz), with a bandwidth of ±2 Hz, also implemented using an FIR filter.

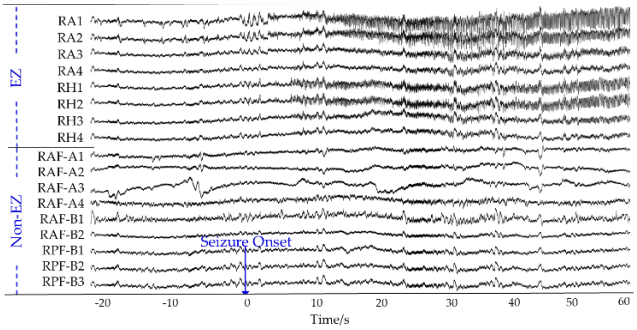


Fig. 2. An exemplary sEEG segment of patient HUP116 during the first seizure.

B. *Dynamic Evolution of Brain Effective Connectivity Networks*

Brain effective connectivity networks were constructed using patient sEEG data. We analyzed the first seizure of patient HUP116 as an example, the sEEG recording had a total duration of 322 seconds, comprising 160,000 data points. The seizure onset occurred at 120 seconds and ended at 261.998 seconds. For analysis, a 160-second segment spanning 80 seconds before and after seizure onset (i.e., 40–200 seconds) was extracted.

To characterize the dynamic evolution of brain effective connectivity during this period, the signals were divided into 5-second time windows with 80% overlap for sliding-window analysis. Within each window, a nonlinear Granger causality model with sparse coefficient regularization was applied to construct a 50×50 causal connectivity matrix. Each element in the matrix, located at row i and column j, represents the causal influence of channel i on channel j within the corresponding time window.

A total of 156 causal connectivity matrices were obtained from the entire sEEG segment, with each matrix representing the effective connectivity network during a specific time window. Fig. 3 schematically illustrates the network structures derived from different representative time windows, depicting different brain network states before and during ictal.

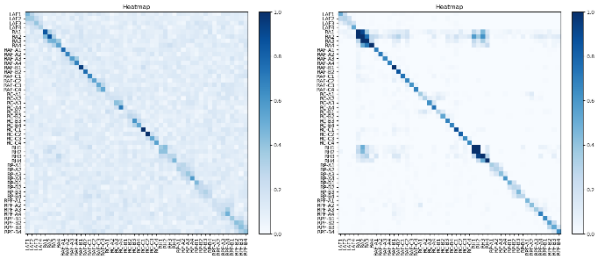


Fig. 3. Brain network states at different stages. The left and right panel correspond to pre-ictal and ictal time windows, respectively.

As shown in Fig. 3, the greater the influence of a certain channel on other channels in the network, the stronger its causal correlation with other channel nodes, and the more distinct the heatmap color of its corresponding row. The results indicate that, except for cases where adjacent region channels have higher connection strength, the connection strength of EZ channels corresponding to RA and RH is significantly higher than that of other channel nodes.

The outflow, inflow, and net flow of each channel were calculated for each time window, and the corresponding flow chart was plotted accordingly (see Fig. 4). The horizontal axis represents time, with a coordinate value of 0 representing the moment of seizure onset. The vertical axis represents the flow magnitude of each channel. Each red and blue line corresponds to an epileptogenic channel and a non-epileptogenic channel, respectively. The gray shaded area indicates the ictal period. The actually labeled epileptogenic channels showed significantly higher net outflow than other channels after the seizure.

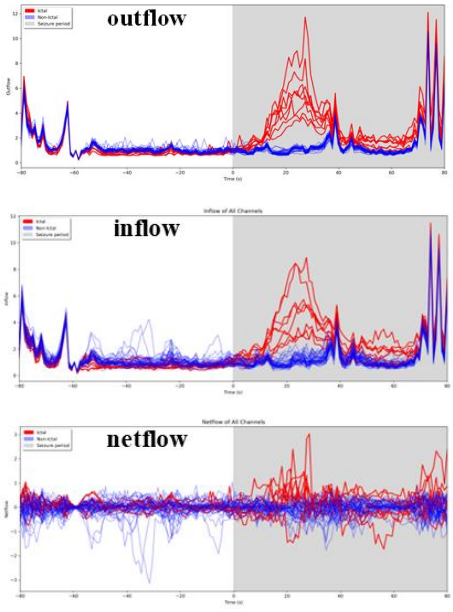


Fig. 4. Color-coded flow diagrams of all channels for each time window. The red channels represent clinically labeled EZ channels, the blue channels represent non-epileptogenic channels, and the gray shaded areas indicate the periods of ictal.

The dynamic evolution of the brain effective connectivity network is illustrated in Fig. 4. Prior to seizure onset, the outflows and inflows of the majority of brain region are

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relatively balanced, resulting in net information flow values fluctuating around zero, which reflects a relatively stable state of inter-regional information transfer. In contrast, during seizure activity, the dynamic characteristics of epileptogenic zone channels are markedly distinct from those of normal regions, exhibiting significantly elevated outflow and inflow, as well as larger net flow amplitudes compared to other channels. These findings indicate that the epileptogenic zones act as core drivers within the seizure network, providing a reliable indicator for seizure localization.

C. Localization of the Epileptogenic Zone

Guided by the characteristics of the effective connection network, this study further compared the characteristics of distinct channels in each time window, and chose the candidate channels that were significantly active in the distribution of information flow. If the channels were active in the continuous time window, they could be considered to be in the core position in the epilepsy network and determined as epileptogenic foci, so as to avoid the interference of sporadic fluctuations. Specifically, three indicators of outflow, inflow and net flow were used to analyze each channel, and the top 18% nodes were selected as the predicted epileptogenic foci. The predicted results were compared with the surgical resection area of epilepsy patients, and the accuracy, sensitivity and F1 Score of the predicted results were calculated.

The statistical metrics used to evaluate classification performance in this approach include Precision, Sensitivity, and F1 score, which are formally defined as follows:

$$Precision = \frac{TP}{TP+FP} \times 100\% \tag{11}$$

$$Sensitivity = \frac{TP}{TP+FN} \times 100\% \tag{12}$$

$$F1\ Score = \frac{2\times(Precision\times Sensitivity)}{Precision+Sensitivity} \tag{13}$$

among them, TP (True Positive) represents the channel that is correctly predicted to be an epileptic source channel and is consistent with the actual surgical resection or clinical labeling.; TN (True Negative) represents the channels correctly identified as non-epileptic sources; FP (False Positive) represents non-epileptic source channels that are wrongly predicted to be epileptic sources and FN (False Negative) indicates an epileptic source channel that has not been correctly identified.

The results of the experiments are shown in Table I, where bold fonts mark the indicators for which patients perform optimally among the three features. From the overall results, outflow was better than inflow and net flow in all predictors (precision=0.889, sensitivity=1.0, F1=0.941), showing higher stability and reliability. Therefore, outflow can be considered a superior indicator for seizure onset zone localization, more effectively reflecting the core role of epileptogenic regions within the brain network.

TABLE I. RESULTS OF EPILEPTOGENIC ZONE LOCALIZATION WITH THREE METRICS

Metrics	*Precision*	*Sensitivity*	*F1 Score*
Outflow	**88.9%**	**100%**	**0.941**
Inflow	77.8%	87.5%	0.824
Netflow	33.3%	37.5%	0.353

To further validate the effectiveness of introducing sparsity regularization in constructing the brain effective connectivity network, an ablation experiment was conducted. Specifically, a model without sparsity constraint was used as the control group and compared with the sparsity-constrained model (see results in Table 1). Both experiments used the same dataset and processing pipeline, and the performance was evaluated using Precision, Sensitivity, and F1 Score as metrics. The experimental results are presented in Table 2.

TABLE II. RESULTS OF EPILEPTOGENIC ZONE LOCALIZATION WITH THE MODEL WITHOUT SPARSITY CONSTRAINT

Metrics	*Precision*	*Sensitivity*	*F1 Score*
Outflow	77.8%	87.5%	0.824
Inflow	55.6%	62.5%	0.588
Netflow	33.3%	37.5%	0.353

Table 2 presents the localization results of the model without sparsity constraint across the three features. As shown, the performance of this model is consistently inferior to that of the sparsity-constrained model in all evaluation metrics. Specifically, under the outflow feature, the F1 Score decreases from 0.941 to 0.824, and the inflow feature also exhibits varying degrees of decline. These results indicate that introducing sparsity regularization in the construction of the effective connectivity network effectively suppresses redundant connections and highlights core lesion-related channels, thereby improving localization accuracy and stability.

IV. CONCLUSIONS AND FUTURE WORK

The study proposes a nonlinear Granger causality learning method to construct an effective connectivity network via component-wise MLP for iEEG-based EZ localization, aiming to provide help for preoperative localization of the EZ among patients with drug-resistant epilepsy. On the HUP iEEG dataset, the accuracy of the approach was 88.9%, the sensitivity was 100% and the F1 score was 0.941. By integrating machine learning with neural data analytics, this approach can accurately identify the EZ before surgery. The experimental results demonstrate that this approach has good performance in the localization of EZ. In view of the large differences in etiology and seizure types among different patients, the criteria for determining EZ should be adapted to individual data characteristics. In future work, we plan to conduct experiments using a larger patient iEEG dataset to further verify the feasibility of the proposed approach in different types of epilepsy, refine the implementation of optimization techniques in engineering and perform systematic comparative analyses using state-of-the-art methods.

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